Process: Enode 2 EDP Integration

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Overview

This document guides the integration of EDP with the SEL platform for the **EV&PV4home Part 2** project. It includes guidelines for the control of EV and informational Webhook subscriptions.

Integration API

The API system for data exchange, payload structure, and endpoints are presented in this section. The payload structure builds on the SEL base payload structure and a sample payload (provided by EDP) is adapted as an example. The variable names in the example also follow SEL's internal nomenclature.

SEL Base Payload

Request Payload

The general base payload format is shown below:

```
[
    {
        "ver": "0.0",
        "g_scope": "vehicle",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740418221,
        "g_id": {"key1": "value1", "key2": "value2"},
        "props": [
            {
                "l_scope": "charging",
                "l_id": {"key1":"value1", "key2": "value2"},
                "data": [
                    {"key": "key1", "value": "var", "unit": "", ...},
                    {"key": "key2", "value": ["min", "max"]},
                    {"key": "key3", "value": ["from", "to"]},
                    {"key": "key3", "value": ["start", "end"]},
                    {"key": "key3", "value": ["min", "avg", "max"]}
                ]
            }
        ]
    }
]

```

Response Payload

The general base payload format is shown below:

```
[
    {
```

```
        "ver": "0.0",
        "g_scope": "vehicle",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740418221,
        "g_id": {"key1": "value1", "key2": "value2"},
        "status": {"state": "...", "msg": "...", "code": "..."},
        "props": [
            {
                "l_scope": "",
                "l_id": {"key1":"value1", "key2": "value2"},
                "data": [
                    {"key": "key1", "value": "var", "unit": ""},
                    {"key": "key2", "value": ["min", "max"]},
                    {"key": "key3", "value": ["from", "to"]},
                    {"key": "key3", "value": ["start", "end"]},
                    {"key": "key3", "value": ["min", "avg", "max"]}
                ]
            }
        ]
    }
]

```

Webhook Payload

The following is the base schema for Webhook `POST` payloads as they will be sent to subscribed listeners:

```
[
    {
        "ver": "0.0",
        "g_scope": "vehicle",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740418221,
        "g_id": {"key1": "value1", "key2": "value2"},
        "props": [
            {
                "l_scope": "",
                "l_id": {"key1":"value1", "key2": "value2"},
                "data": [
                    {"key": "key1", "value": "var", "unit": ""},
                    {"key": "key2", "value": ["min", "max"]},
                    {"key": "key3", "value": ["from", "to"]},
                    {"key": "key3", "value": ["start", "end"]},
                    {"key": "key3", "value": ["min", "avg", "max"]}
                ]
            }
        ]
    }
]

```

Description of Parameters

Parameter	Data Type	Default	Description	Required
ver	*str*	0.0	*Payload version for version control.*	KEY REQUIRED

g_scope (global_scope)	*str*	“”	*Identifies the general scope of the payload. It helps to define the hierarchy in relation.*	**KEY REQUIRED**
msg_id	*str*		*ID to identify requests. Ideally, be unique for every request. It will be included in response.*	**KEY REQUIRED** **VALUE REQUIRED**
tz	*int*		*UTC value of Unix timestamp when the request is sent. For example, 1739109188.*	**KEY REQUIRED** **VALUE REQUIRED**
g_id (global_id)	*dict*	{}	*Key-value pairs to globally identify all entries in the props. For example, a house ID to identify all the assets in the house listed in the prop*	
props	*list*	[]	*List of entries in the request group.*	
l_scope (local_scope)	*str*	“”	*Identifies the local scope of the payload. It helps to define the lowest hierarchy in relation.*	**KEY REQUIRED**
l_id (local_id)	*dict*	{}	*Key-value pairs to locally identify the entry. The l_id and g_id are combined to uniquely identify the entry.*	
data	*list*	[]	*List of parameters of the prop entry. Each entry must have a key field and a value field. Then other metadata in the key-value pair can be added. For example,* • *{“key”: “command”, “value”:”start”)* • *{“key”: “voltage”, “value”: 20.0, “unit”:”V”}*	
key	*str*		*Key name of the parameter. Usually aligns with internal nomenclature.*	**KEY REQUIRED** **VALUE REQUIRED**
value	*any*		*Corresponding value of the parameter. The data type can vary. The value can be as a single value, or list (with 2 or 3 items). If it is a 2-item list, the first value is the starting/min/lowest and the last value is the ending/max/highest. If the 3-item, the first value is the same as the 2-item, the last value is also as the last of the 2-item but the middle value is an average. For example:* • *“Start”* • *[19, 18]*	**KEY REQUIRED** **VALUE REQUIRED**

			• *[20, 24, 28]*	
status	*dict*		*Key-value pairs to identify the status of the request*	
state	*str*		*Represents the current state of the request or process. For example*: • `PENDING` → *The request has been received but has not started processing yet.* • `PROCESSING` → *The request is actively being processed.* • `CONCLUDED` → *The request has been successfully completed.* • `ERROR` → *There was an issue processing the request, and it was not completed.* • `CANCELLED` → *The request was intentionally cancelled before completion.*	KEY REQUIRED VALUE REQUIRED
msg	*str*		*A human-readable message providing additional details about the request status.*	
code	*str*		*A machine-readable status code associated with the request.*	

Authentication

Authentication is based on JSON Web Tokens and is managed using the following two endpoints.

In this context, all requests to the `/ev` endpoint must be authenticated by including the following headers:

```
Content-Type: application/json
Authorization: Bearer <access>
```

Replace **<access>** with the value of `access` returned from an authentication request.

EDP will be provided with generated access credentials for this use case that can use the full capacity of the `/ev` endpoint.

001 Auth: Obtain Token

Description: Get an access token based on provided credentials.

State: TO BE VALIDATED

Endpoint: `POST /token`

Payload:

```
{
    "email": "<email>",
    "password": "<password>"
}
```

Response:

```
{
    "refresh": "<jwt>",
    "access": "<jwt>"
}
```

⚠ A given implementation must bear in mind that access tokens have a short lifespan by default and should include logic to request a new one using request (002).

002 Auth: Refresh Token

Description: Get a new access token using a valid refresh token.

State: TO BE VALIDATED

Endpoint: `POST /token/refresh`

Payload:

```
{
    "refresh": "<refresh>"
}
```

Response:

```
{
    "refresh": "<jwt>",
    "access": "<jwt>"
}
```

Requests

WORK IN PROGRESS

The following is a list of pre-constructed requests to act on specific aspects of the system using the provided base schema.

101 EV: Activate Charge Control

Description: Activate external control of a vehicle's charge state. This enables sending further charge state control requests (102, 103).

ℹ Used to take over the control of the charge state - **grid_solar**

State: TO BE VALIDATED

Endpoint: `POST /ev`

Payload:

```
[
    {
        "ver": "0.0",
        "g_scope": "house",
```

```
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740420930,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d"
        },
        "props": [
            {
                "l_scope": "charging",
                "l_id": { "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1" },
                "data": [
                    { "key": "control", "value": true }
                ]
            }
        ]
    }
]
```

Example Response:

```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740418221,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d",
            "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1"
        },
        "status": {
            "state": "PENDING",
            "msg": "The request has been received but has not started processing yet.",
            "code": "200"
        }
    }
]
```

102 EV: Start Charging

Description: Order the vehicle to start charging. Only works if external control is enabled with request (101).

State: TO BE VALIDATED

Endpoint: `POST /ev`

Payload:

```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740420930,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d"
        },
        "props": [
            {
```

```
                "l_scope": "charging",
                "l_id": { "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1" },
                "data": [
                    { "key": "action", "value": "START" }
                ]
            }
        ]
    }
]
```

Example Response:

```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740418221,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d",
            "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1"
        },
        "status": {
            "state": "PENDING",
            "msg": "The request has been received but has not started processing yet.",
            "code": "200"
        }
    }
]
```

103 EV: Stop Charging

Description: Order the vehicle to stop charging. Only works if external control is enabled with request (101).

State: TO BE VALIDATED

Endpoint: `POST /ev`

Payload:

```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740420930,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d"
        },
        "props": [
            {
                "l_scope": "charging",
                "l_id": { "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1" },
                "data": [
                    { "key": "action", "value": "STOP" }
                ]
            }
        ]
```

```
    }
]
```

Example Response:

```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740418221,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d",
            "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1"
        },
        "status": {
            "state": "PENDING",
            "msg": "The request has been received but has not started processing yet.",
            "code": "200"
        }
    }
]
```

104 EV: Deactivate Charge Control

Description: Deactivate external control of a vehicle's charge state. This enables it to be controlled again by Enode's algorithm (if mode is configured for such) and prevents manual requests (102, 103).

Used to return control to regular functionality - **grid** or **off**

State: TO BE VALIDATED

Endpoint: `POST /ev`

Payload:

```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740420930,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d"
        },
        "props": [
            {
                "l_scope": "charging",
                "l_id": { "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1" },
                "data": [
                    { "key": "control", "value": false }
                ]
            }
        ]
    }
]
```

Example Response:

```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740418221,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d",
            "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1"
        },
        "status": {
            "state": "PENDING",
            "msg": "The request has been received but has not started processing yet.",
            "code": "200"
        }
    }
]
```

105 EV: Push Solar Plan

Description: Push a new solar plan (if the ID does not yet exist in our database) OR update information regarding a specific solar plan.

State: TO BE VALIDATED

Endpoint: `POST /ev`

Payload:

```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740420930,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d"
        },
        "props": [
            {
                "l_scope": "solar_plan",
                "l_id": {
                    "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1",
                    "plan_id": "a8f8031b-d914-4308-99e4-b5446db87ed1"
                },
                "data": [
                    { "key": "control", "value": false }
                ]
            }
        ]
    }
]
```

Inside of **data**, the details of the solar plan can be expanded.

Example Response:


```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740418221,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d",
            "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1"
        },
        "status": {
            "state": "CONCLUDED",
            "msg": "The request has been successfully completed.",
            "code": "200"
        }
    }
]
```


Payload Adaptation

WORK IN PROGRESS

This section will provide examples of projected payload structures and how they convert to the presented schemas.

Webhook Payload

Based on the suggested payload by EDP:


```
{
    "timestamp": 1740420930000,
    "houseId": "175d85aa-6e32-47bb-8e4a-b93094288d5d",
    "locationId": "02a3c49f-4cfe-40df-9491-a76da1008a27",
    "userId": "cd40471d-9408-4cf9-bb54-0594335ff7ad",
    "mode": "grid",
    "isSOCGuaranteedByEDP": false,
    "SOCGuaranteedClientConfigured": 10.5
}
```


The following webhook payload is produced, with all fields being included at all times as requested:

Note: property names were simply converted into snake_case to adhere to our in-house schema, but they can also be simplified if needed.

State: **TO BE VALIDATED**

Endpoint: `POST {EDP_URL}`


```
[
    {
        "ver": "0.0",
        "g_scope": "house",
        "msg_id": "25fd2b71-03b4-4e7d-8b7e-8292a6708b4f",
        "tz": 1740420930,
        "g_id": {
            "house_id": "175d85aa-6e32-47bb-8e4a-b93094288d5d",
            "user_id": "cd40471d-9408-4cf9-bb54-0594335ff7ad"
```

```
        },
        "props": [
            {
                "l_scope": "charging",
                "l_id": {
                    "location_id": "02a3c49f-4cfe-40df-9491-a76da1008a27",
                    "vehicle_id": "2a9291e6-ecb3-4eb5-96f2-c01e99bc7ba1"
                },
                "data": [
                    {"key": "mode", "value": "grid"},
                    {"key": "is_soc_guaranteed_by_edp", "value": false},
                    {"key": "soc_guaranteed_client_configured", "value": 10.5}
                ]
            }
        ]
    }
]

```